

VeloSubspace: Phase-Aware Training-Free Reconstruction for Highly Accelerated 4D Flow MRI

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Abstract. CMRx4DFlow2026 Regular Task 1 evaluates accurate 4D Flow MRI reconstruction at $10\times$ – $50\times$ acceleration, where image magnitude must be recovered without distorting the relative phase that encodes velocity. We propose VeloSubspace, a training-free alternating reconstruction with a temporal basis shared across four velocity encodings, magnitude-weighted circular regularization of relative phase, and explicit multi-coil data consistency. All parameters are selected on the fully sampled P005 demo case and then frozen. Evaluation uses the disjoint P006 case at five acceleration factors, with three deterministic masks per rate. The comparison includes zero-filled SENSE, a matched classical Global LR-DC baseline, and the official learned FlowVN reference. VeloSubspace reduces average magnitude nRMSE from 0.293 to 0.193 and velocity angular error from 64.53 to 50.62 degrees relative to zero filling. Against Global LR-DC (0.232 and 59.08 degrees), it improves all four organizer metrics in 15/15 paired masks; the advantage holds at every acceleration and in leave-one-rate-out summaries. With iteration count matched, circular phase regularization lowers velocity relative and angular errors by 0.0476 and 4.58° on average across five fixed masks. The effect is consistent across all three tested $R = 20$ masks. A three-step acquired-sample update, selected on P005 and then frozen, also improves all four FlowVN metrics at every tested rate. All seven second-center/second-vendor deployment runs produce finite outputs, and every final update lowers the acquired-sample residual. VeloSubspace provides a phase-aware, reproducible Task 1 reconstruction without learned weights.

Keywords: 4D Flow MRI · MRI reconstruction · low-rank subspace
· relative phase · data consistency

1 Introduction

Four-dimensional flow cardiovascular magnetic resonance (4D Flow MRI) measures time-resolved three-directional blood velocity throughout a 3D volume. Its clinical reliability depends on complex images: magnitude defines signal support,

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while relative phase among velocity encodings determines the velocity vector. The spatial, cardiac, coil, and encoding dimensions also make acquisition long, motivating aggressive acceleration [1].

Compressed sensing exploits incoherent sampling and transform sparsity [5]; dynamic MRI also admits compact temporal representations and low-rank-plus-sparse decompositions [6]. For 4D Flow, Valvano et al. combined low-rank and Hadamard sparsity across velocity encodings [8]. Learned unrolled models encode stronger reconstruction priors. FlowVN demonstrated rapid 4D Flow reconstruction [9], its recent cross-site study examined transfer across acquisition settings [7], and FlowMRI-Net studied self-supervised complex-valued unrolling across aortic and cerebrovascular data [4].

The CMRx4DFlow2026 Regular Task 1 scores magnitude SSIM and normalized root mean squared error (nRMSE), together with velocity-magnitude relative error and velocity angular error, at $10\times$ – $50\times$ acceleration [2]. For this task, we propose VeloSubspace, a compact training-free method that preserves measured samples and regularizes the velocity-encoding phase on its natural circular domain. Zero-filled SENSE quantifies the undersampling penalty. Global LR-DC is the principal classical comparator: it uses the same temporal projection and data-consistency machinery but omits the proposed phase term. FlowVN serves as the official learned reference. The paper makes four contributions:

1. a joint complex temporal projection, relative-phase circular regularizer, and multi-coil consistency update;
2. a leakage-controlled benchmark against zero-filled SENSE and a standard global low-rank reconstruction, including acceleration-stratified, worst-pair, and leave-one-rate-out analyses;
3. the official FlowVN curve, a P005-frozen measured-sample refinement, and real-case regional diagnostics of high-speed and boundary attenuation; and
4. a prespecified five-case target-free audit plus two exploratory subject replications from a second center and scanner vendor, with explicit separation between physics-consistency evidence and reference-based accuracy.

Repeated masks are treated as correlated reconstructions of the same subject, not as independent subjects.

2 Method

2.1 Forward model and initialization

Let $x_{v,t} \in \mathbb{C}^N$ be the image for velocity encoding v and cardiac phase t , S_c the sensitivity of coil c , F the 3D Fourier transform, and M_t a time-varying Cartesian mask. Measurements are

$$y_{v,t,c} = M_t F(S_c x_{v,t}) + \eta_{v,t,c}. \quad (1)$$

After normalizing sensitivities so that $\sum_c |S_c|^2 \simeq 1$, the zero-filled SENSE estimate is $x_{v,t}^{(0)} = \sum_c \bar{S}_c F^{-1}(y_{v,t,c})$.

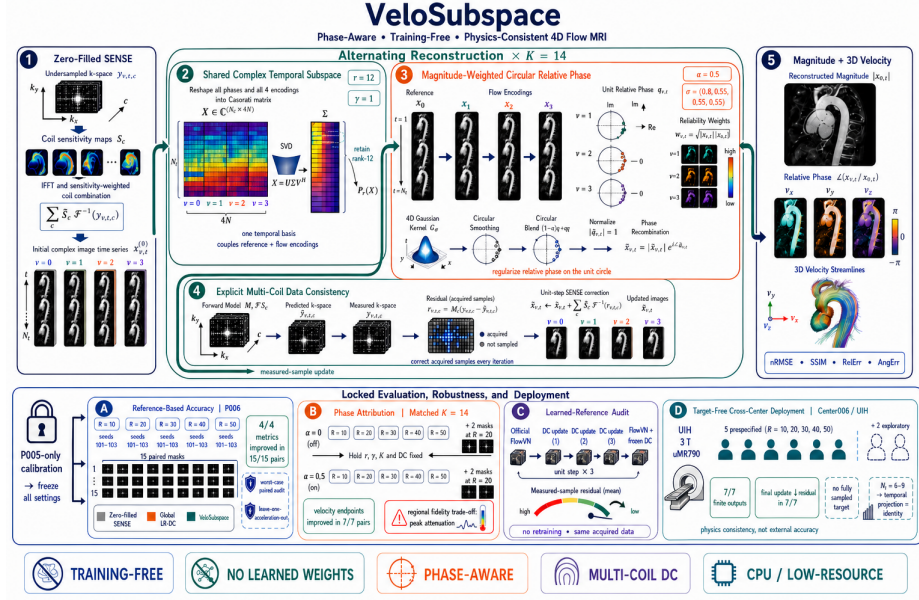


Fig. 1. VeloSubspace workflow and evaluation design. Four velocity encodings share a temporal basis; magnitude-weighted circular relative-phase regularization and explicit multi-coil consistency alternate for 14 frozen iterations. The bottom row separates P006 reference-based evaluation from the target-free Center006/UIH deployment audit.

2.2 Joint temporal projection

All cardiac phases and four velocity encodings are arranged in a Casorati matrix $X \in \mathbb{C}^{N_t \times 4N}$. If $X = U \Sigma V^H$, the rank- r projection is

$$\mathcal{P}_r(X) = U(:, 1:r) \Sigma(1:r, 1:r) V(:, 1:r)^H, \quad \tilde{X} = (1 - \gamma)X + \gamma \mathcal{P}_r(X). \quad (2)$$

The shared temporal basis couples reference and flow encodings. Parameter γ controls how strongly the projection attenuates dynamics.

2.3 Circular relative-phase regularization

For $v = 1, 2, 3$, define the unit relative phasor and reliability weight

$$q_{v,t} = \frac{\tilde{x}_{v,t} \tilde{x}_{0,t}^*}{|\tilde{x}_{v,t} \tilde{x}_{0,t}^*| + \epsilon}, \quad w_{v,t} = \sqrt{|\tilde{x}_{v,t}| |\tilde{x}_{0,t}|}. \quad (3)$$

The circular estimate $q^s = G_\sigma * (wq) / (|G_\sigma * (wq)| + \epsilon)$ is blended with q by weight α , renormalized, and recombined with reconstructed magnitude and reference phase. We fix $\sigma = (0.8, 0.55, 0.55, 0.55)$ over time and space before evaluation.

2.4 Measured-sample update

Every iteration ends with one unit-step SENSE correction:

$$x_{v,t}^+ = \tilde{x}_{v,t} + \sum_c \bar{S}_c F^{-1} [M_t \{y_{v,t,c} - F(S_c \tilde{x}_{v,t})\}]. \quad (4)$$

The temporal, phase, and data-consistency steps are alternated K times. There are no learned weights.

2.5 Calibration and locked comparators

Only P005 is used for calibration. At $R = 20$, seed 17, we minimize magnitude nRMSE plus velocity relative error plus angular error divided by 180° . A sequential search over rank, projection strength, iteration count, and phase weight selects a shared basis with $r = 12, \gamma = 1, K = 14, \alpha = 0.5$. This configuration is frozen before P006. A second P005 mask ($R = 40$, seed 29) is a sensitivity check, not an independent validation subject.

The standard Global LR-DC comparator uses the same shared Casorati projection and Eq. (4), disables phase regularization ($\alpha = 0$), and uses $r = 12, \gamma = 1, K = 3$, selected on the P005 grid. A matched $K = 14, \alpha = 0$ ablation is reported at prespecified seed 101 for all five rates, with two additional $R = 20$ masks, to separate phase regularization from iteration count and test whether its effect persists across acceleration.

For a learned-reference audit, the official FlowVN output initializes $x_{\text{FVN}}^{(0)}$ and Eq. (4) is applied without a low-rank step. P005 $R = 20$, seed 17 selects a unit step and three updates from steps $\{0.25, 0.5, 0.75, 1\}$ and one to three iterations. This boundary optimum is frozen before P006. The locked $R = 20$, seed-101 test is primary; the same update is then applied to the other rates without tuning.

3 Experiments

3.1 Data and split

We use the official CMRx4DFlow2026 demo release and organizer code [3], following the CMRx data lineage [10]. P005 has matrix $(N_v, N_t, N_c, SPE, PE, FE) = (4, 25, 5, 20, 96, 92)$; held-out P006 has $(4, 25, 5, 19, 112, 112)$. Both are aortic acquisitions from Center012 on a 3 T Philips Ingenia, with five coils and VENC 150 cm/s in each direction. The reference-based study thus tests mask and acceleration sensitivity within one acquisition domain.

The external deployment test uses the official Task S1 validation archive: 40 aortic subjects from Center006 acquired on a UIH 3 T uMR790 scanner. This archive contains coil sensitivities, undersampled k -space, masks, segmentations, and metadata, but no fully sampled target. Before reconstruction, we select the lexicographically first subject at each available rate: P010, P002, P001, P003, and P005 for $R = 10, 20, 30, 40, 50$, respectively. We use this second-center subset only for target-free completion, resource, and acquired-sample consistency endpoints.

3.2 Sampling and methods

The official k - t Gaussian generator produces masks at $R \in \{10, 20, 30, 40, 50\}$. P006 uses deterministic seeds 101–103; realized acceleration is recorded. Zero-filled SENSE, Global LR-DC, and VeloSubspace are run on all 15 pairs. The organizer-supplied pretrained FlowVN checkpoint is run at seed 101 for every acceleration using the official low-memory inference path. Its learned prior and training resources differ from those of the two training-free methods. We therefore treat FlowVN as a competitiveness reference, not a matched-compute ablation. FlowVN+DC uses the same masks and measured samples and adds no trained parameters.

3.3 Metrics and regional fidelity

The organizer’s conversion from complex images to magnitude and flow, MSAC polynomial background-phase correction, ROI, and four Task 1 metrics are used unchanged. SSIM uses the organizer’s global normalization and 11^3 window. We report means and standard deviations over masks descriptively; no hypothesis test is performed over repeated masks or frames.

The primary iteration-matched ablation evaluates all four organizer metrics at seed 101 for $R = 10$ –50; seeds 102–103 provide additional replication at $R = 20$. All rates and endpoints are retained. To probe whether circular smoothing attenuates rapid flow locally, phase is converted to cm/s with $v = \phi \text{ VENC} / \pi$. On the locked $R = 20$, seed-101 case, we calculate (i) nRMSE of the ROI mean-speed curve, (ii) bias of robust peak speed, defined as the maximum frame-wise 95th percentile, (iii) vector relative error in the reference top-5% speed voxels, and (iv) vector relative error in a one-voxel inner boundary shell. These are regional fidelity diagnostics, not validated net-flow or wall-shear-stress measurements.

Experiments use Python 3.9, NumPy 2.0, SciPy 1.13, and PyTorch 2.5 on an Apple M3 CPU without MPS/GPU acceleration.

For the external audit, the frozen P005 settings are applied unchanged and cases are processed sequentially with one CPU thread. Prespecified success requires a finite output and a positive reduction from the residual just before the final Eq. (4) update to the final residual in every case. The validation cases contain only 6–9 cardiac frames; hence frozen rank 12 saturates to effective rank N_t and the temporal projection is the identity. This audit probes the circular-phase and data-consistency path, not external temporal compression.

4 Results

Global LR-DC outperforms zero-filled SENSE on every aggregate metric, making it the stronger classical comparator. VeloSubspace then improves all four metrics over Global LR-DC in 15/15 paired masks. Averaged over pairs, its nRMSE, SSIM, velocity relative error, and angular error are 0.193, 0.603, 0.586, and 50.62 degrees.

Table 1. Held-out P006 magnitude nRMSE (mean over three masks; lower is better). Global LR-DC and VeloSubspace are frozen from P005.

Method	$R = 10$	$R = 20$	$R = 30$	$R = 40$	$R = 50$
Zero-filled SENSE	0.261	0.290	0.299	0.303	0.310
Global LR-DC	0.185	0.225	0.241	0.249	0.258
VeloSubspace	0.137	0.183	0.204	0.214	0.226

Table 2. Average over all 15 P006 acceleration-mask pairs. Repeated masks are correlated reconstructions of one subject.

Method	nRMSE ↓	SSIM ↑	RelErr ↓	AngErr (°) ↓
Zero-filled SENSE	0.293	0.212	0.708	64.53
Global LR-DC	0.232	0.452	0.672	59.08
VeloSubspace	0.193	0.603	0.586	50.62

Rate-stratified means remain favorable at every acceleration. Relative to Global LR-DC, pooled nRMSE, velocity relative error, and angular error fall by 16.8%, 12.8%, and 14.3%, while SSIM rises by 0.152. Even at the least favorable rate, the corresponding gains are 12.5%, 7.5%, 13.6%, and 0.142. No individual pair reverses any endpoint; leave-one-rate-out summaries are also favorable (Supplementary Tables S3–S5).

Table 3 reports the learned-reference experiment across all five accelerations. The frozen update improves all four metrics at every rate. At the primary $R = 20$ test, its normalized acquired-sample residual falls from 0.307 to 0.073 (76.1% relative). FlowVN remains the stronger learned reference; VeloSubspace instead offers explicit phase and measured-data control without training.

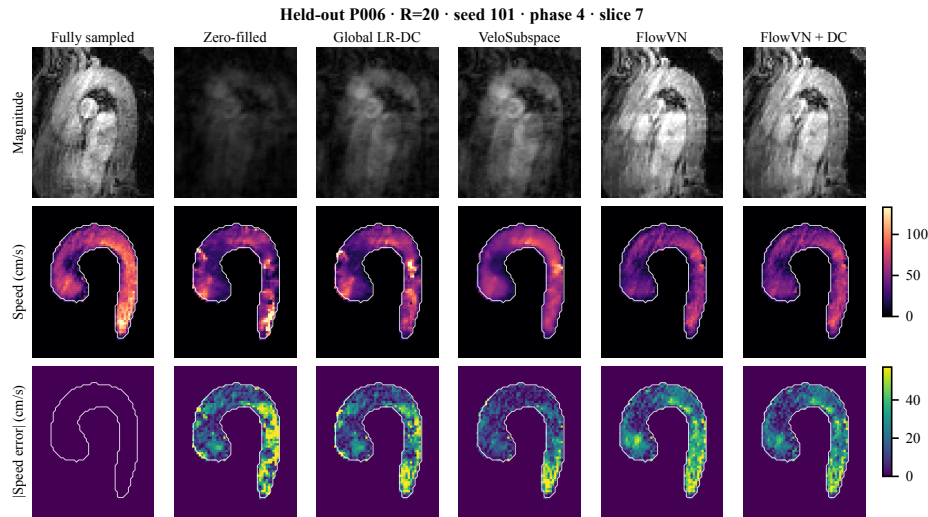
Across the five fixed seed-101 acceleration pairs, relative-phase regularization improves SSIM and both velocity metrics at every rate. Mean paired changes (phase enabled minus disabled) are -0.00005 nRMSE, $+0.00079$ SSIM, -0.0476 velocity relative error, and -4.58° angular error. Magnitude nRMSE improves at three rates and changes by only $+0.00005$ and $+0.00002$ at the other two. The two additional $R = 20$ masks reproduce improvements in SSIM and both velocity endpoints, yielding 7/7 favorable paired velocity comparisons overall.

On the seed-101 regional analysis, relative-phase smoothing lowers the high-speed-voxel and boundary errors from 0.815 and 1.086 to 0.728 and 0.923, respectively. However, its robust peak-speed bias changes from -1.4% to -23.8% , revealing a peak-preservation tradeoff. For FlowVN, the frozen update improves mean-speed-curve nRMSE from 0.259 to 0.205 and peak bias from -31.1% to -24.5% . Its boundary error changes slightly from 0.862 to 0.869.

The frozen pipeline completes all 5/5 external cases with finite outputs. Every final update decreases acquired-sample residual; the median reduction is 4.07% and median final residual is 0.137. Median wall time is 105.3 s per case and

Table 3. Official FlowVN before and after the P005-frozen three-step acquired-sample update on one P006 mask per acceleration. Arrows denote FlowVN $\rightarrow$ FlowVN+DC.

R	nRMSE $\downarrow$	SSIM $\uparrow$	RelErr $\downarrow$	AngErr ($^\circ$) $\downarrow$
10	0.079 $\rightarrow$ 0.068	0.913 $\rightarrow$ 0.937	0.396 $\rightarrow$ 0.344	34.11 $\rightarrow$ 29.35
20	0.122 $\rightarrow$ 0.092	0.877 $\rightarrow$ 0.906	0.462 $\rightarrow$ 0.419	40.15 $\rightarrow$ 38.33
30	0.085 $\rightarrow$ 0.070	0.890 $\rightarrow$ 0.907	0.529 $\rightarrow$ 0.483	45.12 $\rightarrow$ 42.96
40	0.107 $\rightarrow$ 0.104	0.811 $\rightarrow$ 0.817	0.644 $\rightarrow$ 0.618	52.15 $\rightarrow$ 50.45
50	0.099 $\rightarrow$ 0.098	0.836 $\rightarrow$ 0.841	0.605 $\rightarrow$ 0.569	50.69 $\rightarrow$ 48.47

**Fig. 2.** True held-out P006 example at $R = 20$, seed 101, cardiac phase 4, slice 7. FlowVN+DC is included. Slice and frame are selected deterministically from the fully sampled ROI. All methods share display scales; error panels use absolute speed error.

maximum peak resident memory is 1.43 GiB. Median zero-filled-to-final residual reduction is 74.6%.

For an exploratory replication, we run two unused cases selected by small archive footprint: P026 at $R = 20$ and P022 at $R = 40$. Both complete with finite outputs; their final updates reduce residual by 3.49% and 3.32%, respectively. Median zero-filled-to-final reduction is 70.27%. Selection used only rate, non-overlap, and archive footprint before outcomes.

5 Discussion

VeloSubspace improves on both zero-filled SENSE and Global LR-DC. The latter shares the temporal projection and measured-sample update but omits circular phase regularization. The matched- K experiment isolates the phase term while

Table 4. Regional velocity-fidelity diagnostics on the example in Fig. 2. Peak is percent bias; other entries are relative errors.

Method	Mean-curve	Peak (%)	Top-5% jet	Boundary
Zero-filled	0.263	3.5	0.987	1.170
Global LR-DC	0.237	2.6	0.899	1.121
LR-DC matched-14	0.188	-1.4	0.815	1.086
VeloSubspace	0.291	-23.8	0.728	0.923
FlowVN	0.259	-31.1	0.727	0.862
FlowVN+DC	0.205	-24.5	0.722	0.869

Table 5. Prespecified target-free external audit on Center006/UIH. Residuals are normalized on acquired samples; lower is better. The final-step reduction is the prespecified physics-consistency endpoint.

R	10	20	30	40	50
Final residual	0.112	0.145	0.132	0.151	0.137
Final-step reduction (%)	4.59	4.07	2.91	4.22	3.66

holding iteration count fixed. Velocity improves across all seven paired masks, while the regional analysis exposes a peak-speed tradeoff. Rate-stratified, worst-pair, and leave-one-rate-out summaries point in the same direction.

FlowVN addresses a different comparison. It is an official learned reference with access to training-derived priors, rather than a matched-compute ablation. It achieves the strongest image metrics, but its output retains a measurable acquired-sample residual. Three frozen SENSE updates reduce that residual and improve all four organizer metrics at each rate. The experiment shows that an explicit forward-model correction can refine a learned reconstruction without changing its weights.

P005 determines every setting; P006 is reserved for evaluation. An optimum at the edge of the calibration grid may still be anatomy specific. The second P005 mask probes mask sensitivity only. The supplement reports the complete search trajectory.

The five primary and two exploratory Center006/UIH executions test the frozen pipeline beyond the Center012/Philips acquisition domain. Each case produces a finite reconstruction, and every final data-consistency step lowers the measured-sample residual. These are deployment and forward-model checks. The P006 study remains the source of reference-based accuracy. Because the external cases have short cardiac series, rank saturation makes the temporal projection an identity; longer fully sampled series are needed to test cross-center temporal compression.

Limitations. Reference-based accuracy still rests on P006, which shares center, scanner, field strength, anatomy, VENC, and coil count with P005. The seven external executions lack fully sampled targets; their short temporal dimension

also disables low-rank compression at frozen rank. Retrospective masks omit prospective effects, and the organizer ROI is reused for analysis. Regional diagnostics do not validate net flow, regurgitant fraction, pressure, or wall shear stress. Fully sampled external cases remain necessary for a multicenter accuracy table.

6 Conclusion

VeloSubspace combines a shared temporal subspace, circular relative-phase regularization, and explicit measured-sample consistency without neural training. It improves zero filling and Global LR-DC on every P006 mask. Matched ablations across all five rates, together with two further $R = 20$ masks, tie the velocity improvement to the phase-aware update. The same data-consistency step improves the official FlowVN output without retraining. Seven second-center/second-vendor executions finish with finite outputs and lower final-step residuals. VeloSubspace is therefore a reproducible, low-resource reconstruction method for CMRx4DFlow2026 Regular Task 1.

Declarations. This secondary analysis used anonymized challenge data and no new participants; organizer access terms still apply. Code and locked results accompany the submission. The authors declare no external funding or conflicts. AI-assisted language and code tools supported manuscript preparation. The authors verified all analyses, citations, claims, and text and retain full responsibility.

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